Simulations KEY

Name _____

	How you find it	Coin example
Theoretical probability	By reasoning about equally likely outcomes.	$P(\text{heads}) = \frac{1}{2}$
Experimental probability	By doing trials: $\frac{\text{number of times it happened}}{\text{number of trials}}$	20 flips, 12 heads: $\frac{12}{20} = 0.6$

The law of large numbers: the more trials you do, the closer the experimental probability usually gets to the theoretical probability.

A simulation uses random numbers to act out a chance event many times.

1. Assign digits to the outcomes so the probabilities match.
2. Decide what one **trial** is.
3. Run many trials with random digits.
4. Count how often the event happened.

Example: A basketball player makes 70% of her free throws. About how often does she make all 3 of 3 shots?

1. Digits 0 to 6 = make (7 of the 10 digits is 70%). Digits 7, 8, 9 = miss.
2. One trial = 3 digits (3 shots).
3. Use row 1 below. Make 10 trials.

Row 1	**26647 37828 83005 96123 30669 70587**
Row 2	**15021 10728 56722 07313 39118 78635**

Trial	1	2	3	4	5	6	7	8	9	10
Digits	266	473	782	883	005	961	233	066	970	587
Makes	**3**	2	1	1	**3**	2	**3**	**3**	1	1

4 of the 10 trials made all 3. Experimental probability: $\frac{4}{10} = 0.4$.

The theoretical probability is $0.7 \cdot 0.7 \cdot 0.7 = 0.343$. With more trials, the simulation would get closer to it.

Work with a partner. Compare answers when you finish each problem.

1. A spinner has 4 equal parts: red, blue, green, yellow. It is spun 40 times.

Red	Blue	Green	Yellow
14	8	10	8

   (a) Find the experimental probability of red. $\frac{14}{40} = 0.35$

   (b) Find the theoretical probability of red. $\frac{1}{4} = 0.25$

   (c) Why are they different? Chance. 40 spins is not very many trials.

   (d) What would you expect if the spinner were spun 4,000 times? The fraction of reds would be close to 0.25.

2. How would you assign random digits to simulate each event?

   (a) A coin lands heads ($P = 0.5$). 0 to 4 = heads, 5 to 9 = tails

   (b) A student is chosen from a class where 30% walk to school. 0, 1, 2 = walks; 3 to 9 = does not

   (c) A number cube is rolled. 1 to 6 = the number rolled; skip 0, 7, 8, 9

   (d) An event with $P = 0.25$. Use pairs: 00 to 24 = yes, 25 to 99 = no

3. A family has 3 children. Each child is equally likely to be a girl or a boy. Simulate: even digit = girl, odd digit = boy. One trial = 3 digits.

   (a) Use **row 2** on page 1 to run 10 trials. Count the girls in each trial.

Trial	1	2	3	4	5	6	7	8	9	10
Digits	150	211	072	856	722	073	133	911	878	635
Girls	1	1	2	2	2	1	0	0	2	1

   (b) Estimate P(at least 2 girls) from your simulation. $\frac{4}{10} = 0.4$

   (c) The theoretical probability is 0.5. How could you make your estimate closer? Run many more trials.

4. A student flips a coin 50 times and gets 34 heads. She wonders if the coin is fair. (A fair coin lands heads half the time.) A computer simulated 100 sets of 50 flips of a fair coin. The number of heads ranged from 17 to 33. No set had 34 or more heads.

   (a) What does the simulation tell you about getting 34 heads with a fair coin?
   It is very unusual. It did not happen once in 100 simulations.

   (b) Do you think the coin is fair? Explain.
   Probably not. 34 heads would almost never happen with a fair coin, so the result is statistically significant.